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Problems: 1-6

Question 1 - Discuss some everyday examples of uses of supervised learning.

Okay so supervised learning is basically when we have labeled data like input + correct answer and we train the model to map between them. Everyday examples I can think of:

1. *Spam detection in email*: Emails are labeled as “spam” or “not spam.” The model learns patterns from past labeled emails and predicts for new ones
2. *Netflix/Amazon recommendations (partly supervised)*: If users rate movies (like 5 stars, 3 stars etc.), those ratings are labels. The model learns from past user–item–rating data.
3. *Face recognition for unlocking iPhone*: The system is trained on labeled images of your face vs others. It learns to classify: “Is this the authorized user?”
4. *Medical diagnosis systems*: Patient data (symptoms, scans) + labeled diagnosis → model learns to predict disease.

Basically whenever we’re predicting something and we know the “correct answer” during training, that’s supervised learning.

Question 2 - Discuss some every day examples of uses of unsupervised learning.

Here there are no labels, just raw data, and the system tries to find structure. Examples:

1. *Customer segmentation in marketing*: Companies group customers based on buying behavior without pre-labeled categories. The model just clusters similar people together.
2. *Spotify grouping songs*: It might cluster songs into similar “vibes” based on audio features without someone labeling them manually.
3. *Anomaly detection in bank transactions*: The system learns “normal” transaction patterns. If something doesn’t fit, it flags it as unusual without necessarily

having labeled fraud examples.
4. *Google Photos grouping similar faces*: Before you tag anyone, it already clusters similar faces together.

So basically unsupervised learning is like: “Here’s data. Figure something out yourself.”

Question 3 - Discuss some everyday examples of uses of reinforcement learning.

Reinforcement learning is like trial-and-error learning with rewards. Examples:

1. *Self-driving cars*: The car learns actions (accelerate, brake, turn) and gets rewards for safe driving, penalties for collisions.
2. *Game-playing AI (like AlphaGo)*: The agent learns by playing many games and gets reward when it wins.
3. *Ad placement systems*: The system shows different ads and learns which ones get more clicks (reward = clicks).
4. *Robot vacuum cleaners*: They learn better navigation strategies over time by exploring and avoiding obstacles.

Reinforcement learning is basically: Agent + Environment + Actions + Reward ⟶ maximize cumulative reward.

Question 4 - Why should AI/ML systems be able to explain themselves?

1. Trust – If I’m using an AI for medical diagnosis or loan approval, I need to know why it made that decision.
2. Fairness & bias detection – If the model is discriminating based on gender, caste, race, etc., explanation helps detect that.
3. Debugging – Engineers need to know why a model failed.
4. Legal compliance – In many domains (like EU GDPR), people have a “right to explanation.”
5. Safety-critical systems – In autonomous driving or healthcare, blind trust in black-box models is risky.

Basically, if an AI impacts human lives, it can’t just say “trust me bro.”

Question 5 - How can we measure/assess interpretability in ML models?

Interpretability is tricky because it’s not as measurable as accuracy. Some ways:

1. Model-based interpretability

- Simpler models (like linear regression, decision trees) are inherently more interpretable.
- Fewer features → easier to understand.

2. Post-hoc explanation methods

- LIME – explains predictions locally.
- SHAP values – measure feature contribution.
- Feature importance scores
- Saliency maps for images.

3. Human evaluation

- Can humans understand and simulate the model’s prediction?
- User studies measuring trust or comprehension.

4. Faithfulness metrics

- If we remove a feature deemed important, does performance drop?
- Consistency of explanations across similar inputs.

So interpretability is assessed by: Simplicity, Transparency, Fidelity of explanations, Human understanding It’s honestly kind of subjective sometimes.

Question 6 - Are there any other types of learning problems? Examples: Learning to rank Self-supervised learning

a) Self-supervised learning: The model creates its own labels from data. It’s trained on huge unlabeled datasets. Example:

- In NLP (like BERT), mask a word and predict it.
- In vision, predict missing parts of an image.

b) Semi-supervised learning: Uses a small amount of labeled data + lots of unlabeled data. Very practical because labeled data is expensive. Example:

- Medical imaging where only some scans are labeled.

c) Learning to rank: Instead of predicting a class, the model learns to order items based on relevance. Used in:

- Google search results
- Amazon product ranking

d) Online learning: Model updates continuously as new data comes in. Example: stock prediction systems updating with live data.

e) Transfer learning: Using a model trained on one task and adapting it to another. Example: Using ImageNet pretrained CNN and fine-tuning for medical images.